SparseRank: Explainable Cost-Aware Feature Selection for Anomaly Detection in E-Commerce Microservices

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Abstract

Modern e-commerce platforms run on microservice architectures that emits massive stream of system-level signals — CPU, memory, network, disk, response time every second. Collecting, storing, and analysing all of them is expensive as well as it is crucial to identify the root cause in case of abnormal behaviour. We present SparseRank, a lightweight framework that presents which monitoring signals actually matter for anomaly detection in a microservice of E-Commerce business. SparseRank reduces monitoring costs by 94% while maintaining detection quality by combining sparse feature selection with an unsupervised anomaly detector and a feature-attribution layer. SparseRank demonstrates that careful preprocessing, a principled sparsity prior, and a model-agnostic explanation can all work together to provide a useful, deployable observability tool without the need for complex machinery when tested on a real benchmark that covers fourteen fault situations across twelve services.

1 Introduction

Modern e-commerce platforms increasingly depend on microservice architectures-commonly spans ten to twenty loosely coupled services to achieve independent scalability and continuous delivery. Anomaly detection over such systems has been addressed by unsupervised methods. Monitoring costs and explainability remain significant challenges. Recent work on microservice fault localisation focuses on root-cause identification after anomaly confirmation[1]. Our work addresses an earlier, complementary question: how to perform reliable anomaly detection while minimising the number of metrics that must be monitored? We address this through sparse feature selection that is simultaneously performance-conscious and explainable.

2 Literature Review

Explainability for anomaly detection in cloud systems has been addressed by causal methods. Eadro [2] introduces an end-to-end troubleshooting framework using logs, metrics, and traces jointly. While these methods improve detection quality, none addresses monitoring cost and explainability.

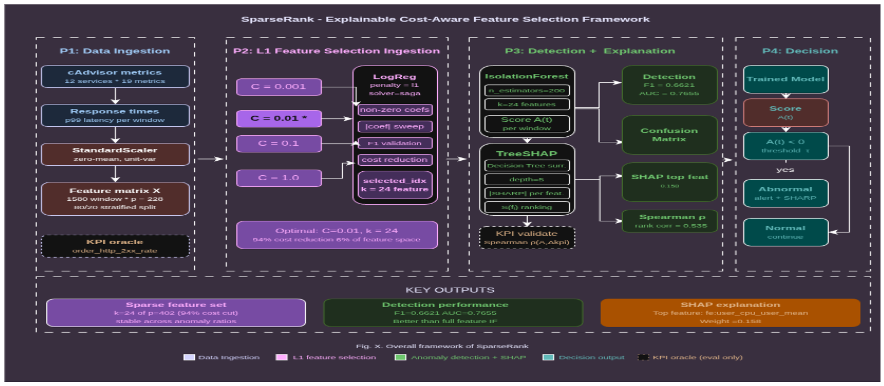

Figure 1: Overall framework of SparseRank.

3 Methodology

This paper presents SparseRank, a three-stage pipeline: (i) L1-regularised logistic regression to select a compact feature subset by penalising redundant metrics to exactly zero; (ii) IsolationForest trained exclusively on the selected features to generate per-window anomaly scores; and (iii) TreeSHAP attribution on a surrogate decision tree to rank selected features by their contribution.

4 Results

SparseRank achieved competitive performance compared with feature-selection baselines while reducing the monitored feature space. SparseRank identifies k = 24 features, resulting in a 94% reduction in monitoring costs, while revealing the top contributing features through SHAP attribution.

5 Conclusion

We have demonstrated that SparseRank is a cost-effective, three-stage anomaly detection method. Ultimately, our proposed framework ensures cost optimisation in observability data monitoring using a dual-ended validation perspective combining dimensionality reduction and explanation-based interpretability.

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